Chapter 9

INTERVAL ESTIMATION AND TESTING OF HYPOTHESES

9.1 Introduction

We studied some methods of estimation of parameters involved in probability distributions and some properties associated with these estimators. Estimators of this type are referred to as *point estimators*. In other words, a single value is suggested as an estimator based on the data. Alternately, one might be interested in proposing an interval or a set in which the parameter is likely to be a member, the interval or the set depending, of course, on the observed data. Such an estimation problem is termed as the problem of interval estimation or the problem of obtaining confidence interval or a confidence set for the parameter. We will explain this concept in detail later in this chapter. Another type of problem that is discussed here is the testing of hypotheses. We might be interested, for instance, to know whether a scalar parameter $\theta \leq \theta_0$ or $\theta > \theta_0$ where θ_0 is a preassigned value. Based on the data on hand, we have to choose one of the two actions. Examples of such problems occur in practice. For instance, suppose a manufacturer of automobiles wants to purchase tyres from a vendor. Suppose the vendor supplies the material in lots of size, say N, some of which are possibly defective. It is obvious that it is impossible to check each and every tyre to determine whether it is defective or not as this process is expensive and time consuming for the manufacturer and the vendor in turn might not allow such an inspection. The manufacturer might be willing to purchase the lot if the percentage defective p of the lot is less than or equal to, say, 5%. Since the exact percentage defective in the lot supplied by the vendor is unknown, the manufacturer has to take a decision, whether to accept or reject the lot supplied by the vendor, on the basis of a sample of size, say, n of tyres from the lot consisting of N tyres. This sort of decision mak-

ing involves errors. The topic of testing of hypotheses discusses different methods of dealing with such problems.

9.2 Interval Estimation (Confidence Interval)

In order to illustrate how to obtain a confidence interval for a parameter of a probability distribution, we will first consider the case of the mean of a normal distribution.

Confidence interval for the mean of a normal distribution

Suppose we are interested in obtaining the information about the unknown parameter θ of a normal population with mean θ and known variance σ^2. Let $X_1, \ldots, X_n$ be a random sample from such a distribution. We have seen that the sample mean $\bar{X}$ is an efficient estimator of θ in the sense that its variance attains the Cramer-Rao lower bound. Hence it is a minimum variance unbiased estimator of θ. It is also a maximum likelihood estimator of θ. Further more

$$Z = \frac{\bar{X} - \theta}{\sigma/\sqrt{n}}$$

has the standard normal distribution when θ is the true parameter. Given $0 < \alpha < 1$, we choose a and b such that

$$P[a \leq Z \leq b] = 1 - \alpha.$$

Since the standard normal distribution is symmetric around 0, let us choose the value $Z_{\alpha/2}$ (see Figure 9.1) such that

$$P[-Z_{\alpha/2} \leq Z \leq Z_{\alpha/2}] = 1 - \alpha.$$

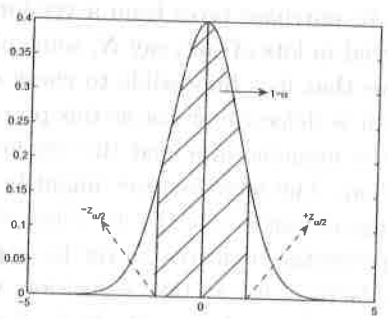

Fig. 9.1

The value $Z_{\alpha/2}$ can be found from the Table C.1 for the standard normal distribution. The above equation can also be written in the form

$$P[-Z_{\alpha/2} \leq \frac{\bar{X} - \theta}{\sigma/\sqrt{n}} \leq Z_{\alpha/2}] = 1 - \alpha$$

when θ is the true parameter . This can again be written in the form

$$P[\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \theta \leq \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}] = 1 - \alpha.$$

Note that the quantity θ is a parameter but not a random variable and the interval $[\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$ has random end points. The end points depend on the observations. How do we interpret such a statement? We are not saying that the conditional probability that the parameter θ belongs to the interval $[\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$ is $1 - \alpha$ given $\bar{X}$. Since the parameter θ is a constant, the probability that it belongs to a specific interval is zero or one since it either belongs to the interval or it does not. What we are saying is that the probability that the random interval contains the parameter θ is $1 - \alpha$. In other words, if the problem of interval estimation is repeated a large number of times, then the relative frequency of the number of intervals, calculated by the above method, containing the parameter θ, approaches the value $1 - \alpha$ in the frequentist interpretation of probability. The random interval

$$[\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}] \tag{9.1}$$

is called a $100(1 - \alpha)\%$ *confidence interval* or *interval estimate* for the parameter θ and the quantity $1 - \alpha$ is called the *confidence coefficient* of the confidence interval. The word confidence comes from the fact that the statistician believes that the random interval is likely to contain the parameter θ about $100(1-\alpha)$ times in repetitions of computing the intervals from 100 i.i.d. samples $\{X_i, 1 \leq i \leq n\}$. Let us now give a formal definition.

Definition. Let $0 \leq \alpha \leq 1$. A random interval $[r_1(\boldsymbol{X}), r_2(\boldsymbol{X})]$ is called a *confidence interval* for a scalar parameter θ with the *confidence coefficient* $(1 - \alpha)$ based on the observations $\boldsymbol{X}$ if

$$P_\theta\{\theta \in [r_1(\boldsymbol{X}), r_2(\boldsymbol{X})]\} = 1 - \alpha$$

for all θ in the parameter space Θ. The interval $[r_1(\boldsymbol{X}), r_2(\boldsymbol{X})]$ is also called an *interval estimator* for the parameter θ with the confidence coefficient $(1 - \alpha)$.

Let us now again consider the normal distribution with an unknown mean θ and an unknown variance σ^2. Suppose we are still interested in obtaining a confidence interval for θ with the confidence coefficient $100(1-\alpha)\%$ based on a random sample $X_1, \ldots, X_n$ from the $N(\theta, \sigma^2)$. Obviously, we cannot use the confidence interval derived earlier for θ as the end points

$$\bar{X} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

for the confidence interval given in (9.1) depend on the unknown parameter σ. We are interested here in information about the mean θ but not about the variance σ^2. The parameter σ is termed as a *nuisance parameter* in this context. Our problem here is to find a confidence interval for θ with a given confidence coefficient in the presence of the nuisance parameter σ^2. From our discussion in the Chapter 5, we know that the statistic t defined by

$$t = \frac{\sqrt{n}(\bar{X}-\theta)}{S}$$

where

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

has the Student's t-distribution with $(n-1)$ degrees of freedom. Recall that S^2 is the sample variance. Note that the statistic t does not depend on the nuisance parameter σ but only on the parameter θ and its distribution function does not depend on either θ or σ. Hence one can use the t-statistic for constructing a confidence interval for the parameter θ even when σ is unknown. Given a constant $0 < \alpha < 1$, we can find values $t_{\alpha/2}$ such that

$$P[-t_{\alpha/2} \le t \le t_{\alpha/2}] = 1-\alpha$$

using the Table C.2 for the t-distribution with $(n-1)$ degrees of freedom. Rewriting the above equation in terms of the parameter θ, we have

$$P[-t_{\alpha/2} \le \frac{\sqrt{n}(\bar{X}-\theta)}{S} \le t_{\alpha/2}] = 1-\alpha,$$

that is

$$P[\bar{X} - t_{\alpha/2}\frac{S}{\sqrt{n}} \le \theta \le \bar{X} + t_{\alpha/2}\frac{S}{\sqrt{n}}] = 1-\alpha.$$

when θ is the true parameter. Hence the random interval

$$[\bar{X} - t_{\alpha/2}\frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}\frac{S}{\sqrt{n}}] \tag{9.2}$$

is a confidence interval for the parameter θ with the confidence coefficient $100(1-\alpha)\%$. Let us compare the confidence intervals obtained in (9.1) and in (9.2). They are essentially similar in form.

Since the t-distribution can be approximated by the standard normal distribution for large n, one can use the interval (9.2) with $t_{\alpha/2}$ replaced by $Z_{\alpha/2}$ for n large. From empirical observations, it is suggested that the sample size n should be at least 30 in order to use the normal approximation.

Let us again consider the confidence interval

$$[\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$$

for the parameter θ of the normal distribution with mean θ and known variance σ^2 with confidence coefficient $100(1-\alpha)\%$. Notice that the length of the interval is

$$2Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

If we want a confidence interval of prescribed length d, then the relation

$$2Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = d$$

should hold or equivalently

$$n = \left(\frac{d}{2Z_{\alpha/2}\sigma}\right)^2. \tag{9.3}$$

This formula gives an approximation to the sample size n needed in order to get a confidence interval of prescribed length d with the prescribed confidence coefficient $100(1-\alpha)\%$. Here σ is assumed to be known. In practice the value of n obtained from equation (9.3) need not be an integer. One chooses the sample size to be the smallest integer greater than or equal to n defined by (9.3).

If the variance σ^2 is unknown but a prior estimate of it is known, then we can use the above relation to get an approximate value of the sample size required. However if σ^2 is unknown, then the length of the confidence interval defined by (9.2) is

$$2t_{\sigma/2}\frac{S}{\sqrt{n}}.$$

It is not possible to determine the constant n from the relation

$$2t_{\alpha/2}\frac{S}{\sqrt{n}} = d$$